

# Laporan Hasil Testing

Report generated on 03-May-2026 at 16:02:48 by [pytest-html](#) v4.2.0

## Environment

|          |                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| Python   | 3.10.0                                                                                                                            |
| Platform | Windows-10-10.0.26200-SP0                                                                                                         |
| Packages | <ul style="list-style-type: none"><li>pytest: 8.3.5</li><li>pluggy: 1.6.0</li></ul>                                               |
| Plugins  | <ul style="list-style-type: none"><li>anyio: 4.13.0</li><li>asyncio: 0.25.3</li><li>html: 4.2.0</li><li>metadata: 3.1.1</li></ul> |

## Summary

158 tests took 00:00:15.

(Un)check the boxes to filter the results.

☒ 0 Failed, ☒ 158 Passed, ☒ 0 Skipped, ☒ 0 Expected failures, ☒ 0 Unexpected passes, ☒ 0 Errors, ☒ 0 Reruns ☒ 0 Retried,

Show all details / Hide all details

| Result | Test                                                                                                         | Deskripsi                                                                            | Duration | Links |
|--------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|-------|
| Passed | tests/chatPipeline_test.py::test_process_query_returns_clarification_when_query_is_ambiguous                 | -                                                                                    | 6 ms     |       |
| Passed | tests/chatPipeline_test.py::test_process_query_returns_security_message_when_sql_validation_fails            | -                                                                                    | 4 ms     |       |
| Passed | tests/chatPipeline_test.py::test_process_query_success_without_visualization                                 | -                                                                                    | 15 ms    |       |
| Passed | tests/chatPipeline_test.py::test_process_query_success_with_visualization                                    | -                                                                                    | 19 ms    |       |
| Passed | tests/chatPipeline_test.py::test_process_query_falls_back_when_analysis_rate_limited                         | -                                                                                    | 5 ms     |       |
| Passed | tests/chatPipeline_test.py::test_process_query_propagates_timeout_http_exception                             | -                                                                                    | 4 ms     |       |
| Passed | tests/chatPipeline_test.py::test_process_query_returns_fallback_message_when_llm_unavailable                 | -                                                                                    | 3 ms     |       |
| Passed | tests/chatStreaming_test.py::test_handle_chat_returns_stream_with_message_words                              | -                                                                                    | 2 ms     |       |
| Passed | tests/chatStreaming_test.py::test_handle_clarification_streams_message_and_keeps_metadata_non_stream         | -                                                                                    | 2 ms     |       |
| Passed | tests/chatbotManagement_test.py::TestChatbotRBAC::test_admin_can_access_chatbot_module                       | Role admin boleh akses module chatbot.                                               | 594 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestChatbotRBAC::test_non_admin_cannot_access_chatbot_module[kepala_divisi] | Role non-admin harus ditolak ketika akses module chatbot.                            | 546 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestChatbotRBAC::test_non_admin_cannot_access_chatbot_module[karyawan]      | Role non-admin harus ditolak ketika akses module chatbot.                            | 546 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestChatbotRBAC::test_chatbot_module_requires_token                         | Tanpa token, endpoint chatbot harus return 401.                                      | 3 ms     |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_kepala_divisi_success                        | POST chatbot kepala_divisi dengan semua field lengkap harus return 201.              | 195 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_karyawan_success                             | POST chatbot Karyawan harus return 201.                                              | 190 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_without_addon_prompt                         | addon_prompt bersifat opsional — boleh null.                                         | 190 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_same_otoritas_deactivates_previous_active    | Create chatbot aktif baru dengan otoritas sama harus nonaktifkan chatbot aktif lama. | 194 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_duplicate_nama_returns_409                   | Nama chatbot yang sudah ada harus return 409 Conflict.                               | 193 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_duplicate_nama_case_insensitive              | Duplikat nama tidak peka huruf besar/kecil.                                          | 193 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_missing_nama_returns_422                     | nama_chatbot wajib — jika tidak ada harus return 422.                                | 189 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_missing_otoritas_returns_422                 | otoritas wajib — jika tidak ada harus return 422.                                    | 188 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_invalid_otoritas_returns_422                 | otoritas hanya boleh 'kepala_divisi' atau 'karyawan'.                                | 189 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_empty_nama_returns_422                       | nama_chatbot tidak boleh string kosong.                                              | 184 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestCreateChatbot::test_create_nama_stripped_whitespace                     | nama_chatbot dengan spasi di ujung harus di-strip.                                   | 193 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestListChatbot::test_list_empty                                            | List chatbot saat DB kosong harus return data= [] dan total=0.                       | 187 ms   |       |
| Passed | tests/chatbotManagement_test.py::TestListChatbot::test_list_returns_all_active                               | Semua chatbot aktif harus muncul di list.                                            | 190 ms   |       |

| Result  | Test                                                                                                                       | Deskripsi                                                                              | Duration | Links |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------|-------|
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_inactive_included                                              | Chatbot dengan is_active=False tetap muncul di list default.                           | 188 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_pagination_limit                                               | limit membatasi jumlah item per halaman.                                               | 193 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_pagination_page_2_different_items                              | Halaman 2 harus return item yang berbeda dari halaman 1.                               | 193 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_invalid_page_returns_422                                       | page < 1 harus return 422.                                                             | 187 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_kepala_divisi                                           | Filter otoritas=Kepala Divisi hanya return chatbot Kepala Divisi.                      | 238 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_karyawan                                                | Filter otoritas=karyawan hanya return chatbot Karyawan.                                | 190 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_invalid_otoritas_returns_422                            | Filter dengan nilai otoritas tidak valid harus return 422.                             | 202 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_search_by_nama                                                 | Search berdasarkan nama_chatbot (partial match).                                       | 206 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_search_by_addon_prompt                                         | Search juga mencakup konten addon_prompt.                                              | 204 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestListChatbot::test_list_search_no_result                                               | Search yang tidak cocok harus return data kosong.                                      | 201 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestGetChatbotById::test_get_by_id_success                                                | GET /{id} harus return detail chatbot yang benar.                                      | 191 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestGetChatbotById::test_get_by_id_not_found                                              | GET dengan ID yang tidak ada harus return 404.                                         | 186 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestGetChatbotById::test_get_inactive_returns_200                                         | Chatbot nonaktif tetap bisa diakses via GET /{id}.                                     | 188 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestGetChatbotById::test_get_invalid_id_returns_422                                       | ID bukan integer harus return 422.                                                     | 186 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_nama_only                                                  | Update hanya nama_chatbot — field lain tidak berubah.                                  | 199 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_otoritas_only                                              | Update hanya otoritas — nama tidak berubah.                                            | 208 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_addon_prompt                                               | Update addon_prompt dengan nilai baru.                                                 | 194 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_addon_prompt_to_null                                       | addon_prompt bisa di-set ke null.                                                      | 189 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_all_fields                                                 | Update semua field sekaligus.                                                          | 189 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_is_active_to_false                                         | is_active bisa di-update melalui PATCH.                                                | 192 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_activate_chatbot_deactivates_other_same_otoritas           | Saat aktivasi chatbot, chatbot aktif lain pada otoritas sama harus dinonaktifkan.      | 207 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_otoritas_active_chatbot_deactivates_active_target_otoritas | Pindah otoritas chatbot aktif harus nonaktifkan chatbot aktif lain di otoritas tujuan. | 203 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_nama_to_existing_returns_409                               | Ganti nama ke nama chatbot lain yang sudah ada harus return 409.                       | 191 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_same_nama_not_conflict                                     | Update ke nama yang sama (milik dirinya sendiri) tidak boleh 409.                      | 194 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_not_found_returns_404                                      | Update chatbot yang tidak ada harus return 404.                                        | 187 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_invalid_otoritas_returns_422                               | otoritas tidak valid saat update harus return 422.                                     | 189 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_empty_nama_returns_422                                     | nama_chatbot tidak boleh string kosong saat update.                                    | 190 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_returns_200                                       | Soft delete harus return 200 dengan pesan sukses.                                      | 190 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_sets_is_active_false                              | Setelah soft delete, is_active di DB harus False.                                      | 192 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_still_accessible_via_get                          | Setelah soft delete, GET /{id} tetap bisa diakses dengan status nonaktif.              | 202 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_still_in_list                                     | Setelah soft delete, chatbot tetap muncul di GET / dengan status nonaktif.             | 197 ms   |       |

| Result  | Test                                                                                                                | Deskripsi                                                                              | Duration | Links |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------|-------|
| Passed                                                                                  | tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_not_found_returns_404                      | Soft delete chatbot yang tidak ada harus return 404.                                   | 197 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_twice_returns_200                          | Soft delete dua kali bersifat idempotent dan tetap return 200.                         | 213 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_returns_200                                | Hard delete harus return 200 dengan pesan sukses.                                      | 199 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_removes_from_db                            | Setelah hard delete, record benar-benar hilang dari DB.                                | 194 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_not_found_returns_404                      | Hard delete chatbot yang tidak ada harus return 404.                                   | 187 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_nama_can_be_reused                         | Setelah hard delete, nama yang sama bisa dipakai lagi.                                 | 197 ms   |       |
| Passed                                                                                  | tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_default_delete_is_soft                                 | Tanpa ?hard=true, default adalah soft delete — record masih ada di DB.                 | 206 ms   |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_detects_ambiguity                          | Test LLM-based detection mendeteksi ambiguitas.                                        | 8 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_clear_query                                | Test LLM menganggap query yang jelas sebagai tidak ambiguous.                          | 11 ms    |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_json_in_markdown_fence                     | Test parser tetap menerima JSON di dalam markdown code fence.                          | 9 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_tie_breaking_rule                          | Test tie-breaking rule: score 0.55-0.65 treated as NOT ambiguous.                      | 9 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_api_error_fallback                         | Test fallback behavior ketika LLM API error.                                           | 10 ms    |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_invalid_json_fallback                      | Test fallback ketika LLM return invalid JSON.                                          | 8 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestClarificationQuestionGenerator::test_generate_question_from_llm           | Test generate pertanyaan dari LLM.                                                     | 9 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestClarificationQuestionGenerator::test_generate_question_llm_error_fallback | Test fallback ke template ketika LLM error.                                            | 9 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestClarificationService::test_process_direct_answer_no_ambiguity             | Test direct answer ketika query tidak ambiguous.                                       | 22 ms    |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestClarificationService::test_process_clarification_needed                   | Test clarification diperlukan ketika query ambiguous.                                  | 22 ms    |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestClarificationService::test_max_clarification_limit                        | Test max clarification limit per session.                                              | 2 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestSessionContextManager::test_create_session_context                        | Test create new session context.                                                       | 0 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestSessionContextManager::test_add_clarification_to_history                  | Test add clarification to session history.                                             | 0 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestSessionContextManager::test_scope_preference_storage                      | Test store scope preference dari clarification answer.                                 | 0 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestSessionContextManager::test_preference_persistence_across_queries         | Test scope preference persist across multiple queries.                                 | 0 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestSessionContextManager::test_session_ttl_cleanup                           | Test session TTL cleanup mechanism.                                                    | 0 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestScenarios::test_scenario_llm_ambiguous_query                              | Skenario: LLM mendeteksi ambiguitas dan generate pertanyaan.                           | 9 ms     |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestScenarios::test_scenario_no_ambiguity                                     | Skenario: LLM menganggap query sudah jelas.                                            | 10 ms    |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestScenarios::test_scenario_tie_breaking                                     | Skenario: Score borderline (0.55-0.65) apply tie-breaking.                             | 16 ms    |       |
| Passed                                                                                  | tests/clarificationMechanism_test.py::TestScenarios::test_scenario_llm_unavailable                                  | Skenario: LLM tidak tersedia, fallback to safe default.                                | 9 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_create_kpi_group_master_with_explicit_sheet_info                                   | Create KPI Group master dengan sheet_id dan nama_grup yang diberikan.                  | 5 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_create_kpi_group_auto_fetch_sheet_info                                             | Create KPI Group auto-fetch sheet_id dan nama dari Google Sheets jika tidak diberikan. | 4 ms     |       |

| Result  | Test                                                                                                | Deskripsi                                                                            | Duration | Links |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|-------|
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_create_kpi_group_tracker                                           | Create KPI Group dengan group_type=tracker.                                          | 4 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_list_kpi_groups_all                                                | List semua KPI Groups dengan pagination.                                             | 4 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_list_kpi_groups_with_filter                                        | List KPI Groups dengan filter group_type.                                            | 3 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_list_kpi_groups_with_search                                        | List KPI Groups dengan search keyword.                                               | 3 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_list_kpi_groups_with_tahun_filter                                  | List KPI Groups dengan filter tahun.                                                 | 5 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_list_kpi_groups_pagination                                         | List KPI Groups dengan pagination calculation.                                       | 4 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_get_kpi_group_by_id                                                | Get satu KPI Group dengan all related records.                                       | 3 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_get_kpi_group_not_found                                            | Get KPI Group dengan ID yang tidak ada.                                              | 3 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_get_kpi_group_master_with_records                                  | Get master KPI Group dengan all master records.                                      | 4 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_update_kpi_group_partial                                           | Update beberapa field pada KPI Group.                                                | 3 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_update_kpi_group_sheet_url_triggers_re_ingestion_master            | Update sheet_url pada master group triggers KPI Master re-ingestion.                 | 5 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_update_kpi_group_tahun_triggers_re_ingestion_master                | Update tahun pada master group triggers KPI Master re-ingestion.                     | 5 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_update_kpi_group_sheet_url_triggers_re_ingestion_tracker           | Update sheet_url pada tracker group triggers Tracker re-ingestion.                   | 5 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_update_kpi_group_master_requires_tahun_when_sheet_url_changes      | Update sheet_url pada master group tanpa tahun akan re-ingest dengan tahun=None.     | 5 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_update_kpi_group_not_found                                         | Update KPI Group dengan ID yang tidak ada.                                           | 3 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_delete_kpi_group                                                   | Delete KPI Group by ID.                                                              | 5 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_delete_kpi_group_commits_transaction                               | Delete operation commits transaction.                                                | 4 ms     |       |
| Passed                                                                                  | tests/kpiGroupCRUD_test.py::test_create_list_get_delete_flow                                        | Complete CRUD flow: create → list → get → delete.                                    | 6 ms     |       |
| Passed                                                                                  | tests/promptTemplate_test.py::test_build_analysis_prompt_supports_datetime_values                   | -                                                                                    | 0 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerRepository::test_get_config_returns_default_config            | get_config selalu mengembalikan config (default jika file belum ada).                | 2 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerRepository::test_update_config_saves_to_manager               | update_config harus memanggil save_config dengan perubahan yang diberikan.           | 2 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerService::test_build_trigger_returns_cron_trigger              | _build_trigger harus menghasilkan CronTrigger.                                       | 1 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerService::test_build_trigger_fires_on_correct_day_and_hour     | CronTrigger harus fire pada hari dan jam yang benar tiap bulan.                      | 1 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerService::test_auto_pause_if_december_pauses_in_december       | _auto_pause_if_december harus remove job dan set is_enabled=False di bulan Desember. | 4 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerService::test_auto_pause_if_december_skips_non_december       | _auto_pause_if_december tidak boleh pause di bulan selain Desember.                  | 2 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerService::test_register_job_enabled_adds_cron_job              | register_job harus menambah CronTrigger job saat config enabled.                     | 2 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerService::test_register_job_disabled_does_not_add_job          | register_job tidak boleh add job jika scheduler dinonaktifkan.                       | 2 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerService::test_register_job_replaces_existing_job              | Jika job lama ada, register_job harus remove lalu add yang baru.                     | 3 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerService::test_get_next_run_time_returns_none_when_job_missing | get_next_run_time harus None ketika job scheduler belum ada.                         | 1 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerController::test_get_config_returns_default                   | get_config harus selalu mengembalikan config (tidak pernah None).                    | 2 ms     |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerController::test_update_config_updates_and_registers_job      | update_config harus simpan perubahan,                                                | 3 ms     |       |

| Result  | Test                                                                                           | Deskripsi                                                              | Duration | Links |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------|-------|
|                                                                                         |                                                                                                | register job, dan isi next_run_at.                                     |          |       |
| Passed                                                                                  | tests/scheduler_test.py::TestSchedulerController::test_trigger_now_runs_ingestion_job          | trigger_now harus mengeksekusi run_ingestion_job.                      | 2 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogin::test_login_sukses                                     | Login valid mengembalikan access token, refresh token, dan data user.  | 5 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogin::test_login_password_salah                             | Login dengan password salah harus ditolak 401.                         | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogin::test_login_user_tidak_ditemukan                       | Login dengan identifier yang tidak terdaftar harus return 401.         | 3 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogin::test_login_akun_nonaktif                              | User nonaktif tidak boleh login dan harus mendapat 403.                | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogin::test_login_via_email                                  | Login via email (bukan username) tetap valid dan mengembalikan 200.    | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogin::test_login_payload_kosong                             | Payload login kosong harus gagal validasi request (422).               | 3 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestRefresh::test_refresh_sukses_rotation                        | Refresh token valid harus merotasi token dan menghasilkan token baru.  | 7 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestRefresh::test_refresh_token_sudah_direvoke                   | Refresh token yang sudah direvoke harus ditolak 401.                   | 5 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestRefresh::test_refresh_token_kadaluarsa                       | Refresh token kadaluarsa harus ditolak 401.                            | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestRefresh::test_refresh_token_palsu                            | Refresh token palsu dengan format JWT tidak valid harus ditolak 401.   | 3 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestRefresh::test_refresh_kirim_access_token                     | Access token dikirim ke /refresh → ditolak karena type mismatch.       | 18 ms    |       |
| Passed                                                                                  | tests/userManagement_test.py::TestRefresh::test_refresh_user_nonaktif                          | Refresh token milik user nonaktif harus ditolak 403.                   | 5 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestRefresh::test_refresh_user_dihapus                           | Refresh token untuk user yang sudah tidak ada harus ditolak 401.       | 6 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestRefresh::test_refresh_revoke_dipanggil_sekali                | Pada refresh sukses, revoke_token harus dipanggil tepat satu kali.     | 6 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogout::test_logout_sukses                                   | Logout valid harus revoke refresh token dan mengembalikan 200.         | 5 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogout::test_logout_token_palsu                              | Logout dengan refresh token palsu harus ditolak 401.                   | 3 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogout::test_logout_token_kadaluarsa                         | Logout dengan refresh token kadaluarsa harus ditolak 401.              | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestLogout::test_logout_idempotent                               | Logout dua kali tidak boleh error (revoke_token bersifat idempotent).  | 6 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestCreateUser::test_create_user_sukses                          | Admin dapat membuat user baru ketika username/email belum terpakai.    | 194 ms   |       |
| Passed                                                                                  | tests/userManagement_test.py::TestCreateUser::test_create_user_duplikat_username               | Pembuatan user harus gagal 409 jika username sudah terdaftar.          | 5 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestCreateUser::test_create_user_duplikat_email                  | Pembuatan user harus gagal 409 jika email sudah terdaftar.             | 5 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestCreateUser::test_create_user_forbidden_sebagai_kepala_divisi | Kepala divisi tidak boleh membuat user baru (POST /users hanya admin). | 3 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestCreateUser::test_create_user_forbidden_sebagai_karyawan      | Karyawan tidak boleh membuat user baru.                                | 3 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestCreateUser::test_create_user_password_lemah                  | Password tanpa huruf kapital → 422.                                    | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_sukses                       | Admin dapat melihat daftar user dengan metadata total yang benar.      | 6 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_pagination                   | Parameter page dan limit harus diteruskan ke repository dengan benar.  | 5 ms     |       |

| Result  | Test                                                                                              | Deskripsi                                                                            | Duration | Links |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------|-------|
| Passed                                                                                  | tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_with_search_role_status_filter  | Parameter search, role, dan status diteruskan ke service/repository.                 | 6 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_forbidden_sebagai_kepala_divisi | Kepala divisi tidak boleh akses list semua user.                                     | 3 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestGetAllUsers::test_get_all_users_forbidden_sebagai_karyawan      | Karyawan tidak boleh akses list semua user.                                          | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestGetUserById::test_get_user_by_id_sukses                         | Admin dapat mengambil detail user berdasarkan user_id yang valid.                    | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestGetUserById::test_get_user_by_id_not_found                      | Pengambilan detail user harus return 404 jika user_id tidak ditemukan.               | 4 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_hash_dan_verify_password                  | hash_password menghasilkan hash bcrypt yang lolos verify_password untuk nilai benar. | 557 ms   |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_create_access_token_berisi_field_wajib    | Access token harus memuat sub, username, role, type access, dan expiry positif.      | 1 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_create_refresh_token_berisi_field_minimal | Refresh token hanya memuat field minimal tanpa role/username.                        | 1 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_decode_access_token_tolak_refresh_token   | decode_access_token harus menolak token bertipe refresh.                             | 2 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_decode_refresh_token_tolak_access_token   | decode_refresh_token harus menolak token bertipe access.                             | 1 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_decode_token_kadaluarsa                   | decode_access_token harus melempar 401 untuk token yang sudah kadaluarsa.            | 8 ms     |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_authenticate_user_sukses                  | authenticate_user mengembalikan user saat identifier dan password valid.             | 14 ms    |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_rotate_tokens_revoke_lama_terbit_baru     | rotate_tokens harus revoke token lama dan menerbitkan pasangan token baru.           | 18 ms    |       |
| Passed                                                                                  | tests/userManagement_test.py::TestAuthServiceUnit::test_rotate_tokens_reuse_ditolak               | rotate_tokens harus menolak refresh token yang terdeteksi reuse/revoked.             | 65 ms    |       |